How-To Library

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Using Regular Expressions to Specify Folders in Unstructured Data Stores

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Abstract

This article explains how to enter regular expressions to specify the folders that Data Privacy Management includes and excludes when scanning unstructured data stores with a remote agent. You can enter regular expressions to specify folders for the following data store types: Amazon S3, HDFS, file system, Google Drive, Microsoft Azure Data Lake, Microsoft Azure Blob Storage, Microsoft OneDrive, and Microsoft SharePoint.

Supported Versions

- Data Privacy Management 10.4.1

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Overview

For unstructured data stores that scan with a remote agent, you can enter regular expressions to specify the folders that the Scan job includes or excludes.

You can enter regular expressions for the following data store types: Amazon S3, Hadoop Distributed File System (HDFS), file system, Google Drive, Microsoft Azure Data Lake, Microsoft Azure Blob Storage, Microsoft OneDrive, and Microsoft SharePoint.

The total length of file paths and regular expressions that you can enter in the **Folder Regular Expression** text field cannot exceed 255 characters. If the total length exceeds 255 characters, you can decrease the number of characters by enhancing regular expressions with the asterisk (`*`) operator.

You can use the following operators in regular expressions:

- Asterisk (`*`)
- `AND`
- `NOT`

Example

The parent folder in a data store contains four subfolders. You want Data Privacy Management to scan three of the subfolders but not the fourth subfolder.

You can enter a regular expression in the following format using the `AND` and `NOT` operators:

```
<parent folder absolute path>/<folder1> AND <parent folder absolute path>/<folder2> AND <parent folder absolute path>/<folder3> NOT <parent folder absolute path>/<folder4>
```

To shorten the expression, you can replace absolute folder paths and folder names with additional regular expressions in the following format, where *regex* indicates a regular expression:

```
<parent folder regex>/<folder1 regex> AND <parent folder regex>/<folder2 regex> AND <parent folder regex>/<folder3 regex> NOT <parent folder regex>/<folder4 regex>
```

Adding a Folder Regular Expression to Unstructured Data Stores

To specify the folders that Data Privacy Management includes and excludes when you scan an unstructured data store with a remote agent, you can add a regular expression in the data store properties..

1. Click **Manage** > **Data Stores**.

   The **Data Stores** workspace appears.

2. Click **New**.

   The **New_Data_Store** page appears.

3. Select the category and data store type, and verify that the **Scan with Remote Agent** check box is selected.

   Additional properties appear.

4. In the **Folder Options** property, select **Use Regular Expressions**.

   The **Folder Regular Expression** property appears.

5. Enter a regular expression for one or more folders that you want to include or exclude.

   The expression cannot exceed 255 characters. You can optimize and reduce the length further using more regular expressions.

Using the Asterisk (*) Operator in Regular Expressions

Use the asterisk (`*`) operator to specify a range of parent directory subfolders that the Scan job includes or excludes.

The following table provides examples of how to use the asterisk (`*`) operator to configure a regular expression for a parent folder with subfolders that include `HR` in the folder names:

Regular Expression	Description
<parent folder>/HR*	The data store scan includes or excludes folders that have names beginning with the letters `HR`. For example, the data store scan includes or excludes files and folders in directories that have names such as `HR`, `HR-Core`, or `HR/Core/Leader`. The Scan job does not filter folders that have names such as `Core-HR`.
<parent folder>/HR/*	Since the asterisk (`*`) operator follows the `/` character, the data store scan includes or excludes subfolders in the `HR` directory. For example, the data store scan includes or excludes files and folders in directories that have names such as `HR/Core/Leader` or `HR/Hiring`. The Scan job does not filter the `HR` folder or the files in the folder.
<parent folder>/*HR	The data store scan includes or excludes folders that have names ending with the letters `HR`. For example, the data store scan includes or excludes files and folders in directories that have names such as `HR`, `Core-HR`, `/home/user/HR`, and `/home/user/Core-HR`.
<parent folder>/*HR*	The data store scan includes or excludes folders and subfolders that have the letters `HR` in the folder name.
<parent folder>/HR/*/Employees	The data store scan includes or excludes files in the `Employees` folder, which is nested in another subfolder of the `HR` directory.
<parent folder>/HR/*/Employees/*	The data store scan includes or excludes files and folders in the `Employees` folder, which is nested in another subfolder of the `HR` directory. The Scan job does not filter files in the `Employees` folder.

Using the AND Operator in Regular Expressions

Use the `AND` operator to add folders that the Scan job includes or excludes.

The following table provides examples of how to use the `AND` operator:

Regular Expression	Description
<*parent folder*>/HR AND <*parent folder*>/IT	The data store scan filters files in the folders `HR` and `IT`.
<*parent folder*>/HR* AND <*parent folder*>/IT*	The data store scan filters files and subfolders in the folders `HR` and `IT`.

Note: Do not insert the `AND` operator at the beginning of a regular expression. The data store scan does not filter folders that are specified in a regular expression that begins with the `AND` operator.

Using the NOT Operator in Regular Expressions

Use the `NOT` operator to subtract a list of folders that the Scan job includes or excludes.

The following table provides examples of how to use the `NOT` operator:

Regular Expression	Description
<*parent folder*>/HR* NOT <*parent folder*>/HRMS	The data store scan filters files and subfolders in the folder `HR` but not files in the folder `HRMS`. The scan filters subfolders in the `HRMS` folder.
<*parent folder*>/HR* NOT <*parent folder*>/HRMS*	The data store scan filters files and subfolders in the folder `HR` but does not filter files or subfolders in the folder `HRMS`.
<*parent folder*>/HR* NOT <*parent folder*>/HRMS/*	The data store scan filters files and subfolders in the folder `HR` and filters only the files in the folder `HRMS`. The scan does not filter subfolders in the folder `HRMS`.
<*parent folder*>/HR* NOT <*parent folder*>/HR/*/Employees	The data store scan filters files and subfolders in the folder `HR`. The scan does not filter folders named `Employees` that are more than one level beneath the `HR` folder.
FolderOptions NOT *FolderOptions/*/*	The data store scan filters files and subfolders in the folder `FolderOptions`. The scan does not filter files and folders that are two or more levels beneath the `FolderOptions` directory. For example, the scan will filter files in directories such as `FolderOptions/HR` and `FolderOptions/MARKETING` but will not filter folders in these directories.
* NOT *FolderOptions*	The data store scan filters files and subfolders in the file path specified in the **Source Directory** property. The scan does not filter files and folders in directories that contain the string `FolderOptions`.

Note: Do not insert the `NOT` operator at the beginning of a regular expression. The data store scan does not filter folders that are specified in a regular expression that begins with the `NOT` operator.

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